Ivan Cabrilo

ivancabrilo.34@gmail.com | LinkedIn | GitHub | personal_website

EDUCATION

University of Rochester — Rochester, New York
Bachelor of Science in Computer Science, Machine Learning track & minor in Business — *August 2022 – May 2026*

- GPA: 3.66/4.00, Schwarz Discover Research Award
- Honors: Whipple Science Scholarship ($12,000/year), Rochester National Grant ($55,000/year), Shelby Davis UWC Award ($30,000/year)
- Courses: End-To-End Deep Learning, Machine Learning Systems for Efficient AI, Computational Intelligence, Automata & Complexity, Computer Organization, Computer Networks, Design of Efficient Algorithms
- Exchange year abroad at Vrije Universiteit Amsterdam 2024-2025
- Teaching Assistant for CSC 172 – Data Structures & Algorithms

EXPERIENCE

IT Advanced Services — Amsterdam, Netherlands
Machine Learning Engineer Intern — *June 2025 – Present*

- Building an ensemble fraud detection model from scratch by combining anomaly detection with semi-supervised learning, where predictions are blended (0.3 Isolation Forest + 0.7 XGBoost)
- Solved the limited-data problem by generating synthetic samples & creating a self-learning pipeline for fraudulent transaction detection
- Designed a custom labeling framework with confidence weighting (ground truth > custom heuristics > anomalies) to fine-tune XGBoost

Event and Classroom Management — Rochester, New York
Desk-Technician — *March 2023 – Present*

- Diagnosing and repairing software & hardware issues on Apple, Dell, Crestron, Owl and Lenovo devices, as well as Xerox and HP printers

Cikom — Podgorica, Montenegro
Data Science Intern — *June 2023 – August 2023*

- Designed a relational database for inventory management, improving update time by 20% through product categorization and MapReduce
- Enhanced MySQL server performance on AWS EC2 by implementing horizontal scaling and sharding, achieving a 17% increase in query speed

Funko Dojo — Glen Rock, New Jersey
Software Engineer Intern — *November 2022 – January 2023*

- Engineered a robust web scraping framework with Python Requests, utilized BeautifulSoup for parsing the HTML data from retailer websites
- Leveraged Amazon Web Services S3 cloud technologies via the boto3 library to securely store & retrieve scraped data for the Funko Dojo SaaS

ENTREPRENEURSHIP

Sorriso.care
Co-Founder & Director — *April 2024 – September 2025*

- Overseeing end-to-end development of Medical Start up in Montenegro from technical and businesss perspective
- Secured **30,000 €** in funding from the Innovation Fund of Montenegro and **7000 €** from Boost Me Up programs
- Developed business strategy and increased organic traffic through SEO with SE ranking, and targeting Google Ads by 254%
- Negotiated 15% revenue-sharing agreemet with a private clinic; current total revenue generated is **117k €**

RESEARCH

Rochester Human Computer Interaction Lab — Rochester, New York
Undergraduate Machine Learning Research Assistant — *January 2023 – Present*
Parkinson's diseas Project (PARK)

- Built user guide for a PARK web app using TypeScript and Google's MediaPipe for Face Landmark Detection APIs
- Applied data mining to extract features for Parkinson's disease using VisionTransformer and Wav2Vec2ForSequenceClassification
- Fine-tuned WaveLM and V-JEPA deep learning models on 300 audio & 500 video patient recordings
- Employed multimodal fusion by integrating audio-visual embeddings, achieving 75% accuracy in classifying Parkinson's disease patients
- Fine-tuned Clinical LLaMA-70B to generate clinical longitudinal summaries of patient visits for neurologist at UofR Medical Center
- Designed an RL pipeline that uses a DOP-based reward model to steer the LLM toward medically useful and complete patient visit summaries

PROJECTS

Semantic Search

- Built an end-to-end semantic search engine for KurzGesagt YouTube channel using ETL pipelines with SentenceTransformers
- Engineered an ETL pipeline to pull videos and transcripts via YouTube API, clean/transform text, and generate embeddings
- Developed and deployed a search API that computes (cosine) similarity scores across titles and transcripts, returning ranked video results via a FastAPI endpoint, containerized the system with Docker, published to Docker Hub, and deployed on AWS Elastic Container Service

Distributed ResNet Scaling Study

- Fine-tuned ResNet-18/34/50/152 on a multi-GPU cluster (NVIDIA RTXs A5000) using PyTorch pipeline parallelism on CIFAR-10 dataset to discover best global batch size and micro-batch count for optimal throughput and convergence

Bouldering Pose Detection

- Developing algorithm to quantify limb/joint overreliance for data-driven climbing insights and improving my climbing skills
- Implemented a computer vision pipeline using MediaPipe for 3D Pose Landmark Detection that extracts data from my bouldering videos
- Filtered low-visibility points (<65%), averaged hand landmarks, and skipped every second frame to boost processing speed by 100%

TECHNICAL SKILLS

- Programming: Python, Java, TypeScript, JavaScript, SQL, Rest API
- Frameworks/Libraries/Platforms: Next.js, Docker, PyTorch, NodeJS, React, Pandas, NumPy, Matplotlib, TensorFlow, AWS, Google Colab
- Certifications: ML by Stanford University, NVIDIA Building Conversational AI Applications, CodePath Web Dev, CITI Human Research